

Aliah University

Odd (Autumn) – Semester Examination (session 2025-26)

(For 2nd Year 3rd Semester 2 yrs MCA)

Paper Name: Design and Analysis of Algorithm

Full Marks: 80

Paper Code: MCAPGPC17

Time: 3 hrs

Group - A

(Answer all questions)

5 x 2 = 10

1. What is the time and space complexity? Explain with suitable examples. [Remembering]
2. Define recursion with an example. [Remembering]
3. What is the difference between Branch & Bound and Backtracking? [Remembering]
4. Define a brute-force algorithm with suitable example. [Remembering]
5. Define approximation ratio of an algorithm with an example. [Remembering]

Group - B

(Answer any 6 questions)

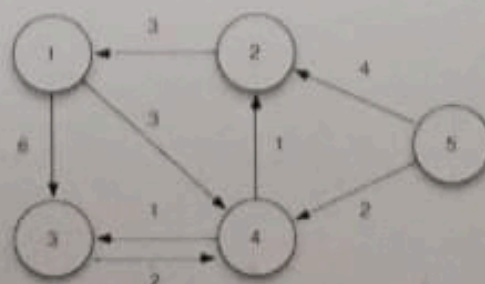
6 x 5 = 30

6. Discuss different types of asymptotic notations with examples. [Remembering] 5
7. Illustrate the working principle of binary search algorithm by taking a suitable example. [Understanding] 5
8. $U = \{1, 2, \dots, 6\}$ and $S' = \{S_1, S_2, \dots, S_{10}\}$. The elements and cost of each set is given below.
 $S_1 = \{1, 2\}$, $S_2 = \{3, 4\}$, $S_3 = \{5, 6\}$, $S_4 = \{1, 2, 3, 4, 5, 6\}$, $S_5 = \{1\}$, $S_6 = \{2\}$, $S_7 = \{3\}$, $S_8 = \{4\}$, $S_9 = \{5\}$, $S_{10} = \{6\}$ and $c(S_1) = 3$, $c(S_2) = 3$, $c(S_3) = 3$, $c(S_4) = 14$, $c(S_5) = 1$, $c(S_6) = 1$, $c(S_7) = 1$, $c(S_8) = 1$, $c(S_9) = 1$, $c(S_{10}) = 1$.
Find an approximation solution for the above set cover problem and calculate its corresponding cost value. [Applying] 5
9. Apply the greedy approach to a machine-scheduling problem where deadlines are involved. The details of the jobs are as follows:

Job	Deadline	Profit
1	2	20
2	4	55
3	1	12
4	3	35

Find the optimal scheduling order of the jobs considering deadline constraints. [Applying] 5

10. Find the single source shortest path for the given graph using Bellman Ford algorithm. [Applying] 5



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11. Given four item with their corresponding weight and profit: A(40, 280), B(10, 100), C(20, 120), D(24, 120). The capacity of the knapsack $W = 70$. Calculate the maximum profit using greedy approach for this knapsack problem. [Applying] 5
12. Describe in detail 4-queens problem using backtracking. [Understanding] 5
13. Compare and contrast DFS and BFS graph traversal algorithms. [Understanding] 5

Group - C

(Answer any 4 questions)

4 x 10 = 40

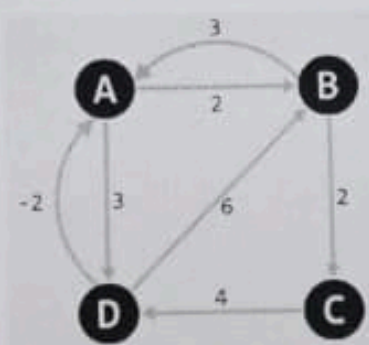
14. Write the algorithm for merge sort and analyse its complexity. Explain it with suitable example. [Understanding] 3+2+5

15. Find an optimal solution (optimal parenthesization) for the matrix chain multiplication problem on the chain of 4 matrices $A_0A_1A_2A_3$, using the dynamic programming approach. The dimensions of the matrices are given below: [Applying] 10

Matrix	A_0	A_1	A_2	A_3
Dimension	(5 x 10)	(10 x 3)	(3 x 12)	(12 x 5)

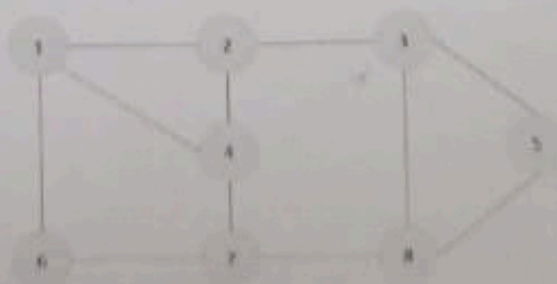
16. Describe the branch-and-bound approach with an example of the 15-puzzle problem. [Applying] 10

17. Solve the all-pair shortest path problem for the following graph using Floyd's Algorithm. [Applying] 10



18. Define P, NP, NP-Hard and NP-complete. Show that clique is NP-complete. [Understanding] 4 + 6

19. Find the approximation solution for vertex cover problem of the given graph and calculate the approximation ratio. [Applying] 10



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